



Bursting microbubbles: How nanobubble contrast agents can enable the future of medical ultrasound molecular imaging and image-guided therapy

Agata A. Exner¹ and Michael C. Kolios²

Abstract

The field of medical ultrasound has undergone a significant evolution since the development of microbubbles as contrast agents. However, because of their size, microbubbles remain in the vasculature and therefore have limited clinical applications. Building a better—and smaller—bubble can expand the applications of contrast-enhanced ultrasound by allowing bubbles to extravasate from blood vessels—creating new opportunities. In this review, we summarize recent research on the formulation and use of nanobubbles (NBs) as imaging agents and as therapeutic vehicles. We discuss the ongoing debates in the field and reluctance to accepting NBs as an acoustically active construct and a potentially impactful clinical tool that can help shape the future of medical ultrasound. We hope that the overview of key experimental and theoretical findings in the NB field presented in this article provides a fundamental framework that will help clarify NB–ultrasound interactions and inspire engagement in the field.

Addresses

¹Departments of Radiology and Biomedical Engineering, Case Western Reserve University, Cleveland, OH, USA

²Department of Physics, Ryerson University, Toronto, ON, Canada

Corresponding authors: Exner, Agata A (agata.exner@case.edu); Kolios, Michael C (mkolios@ryerson.ca)

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Introduction

Biomedical ultrasound (US) imaging is a well-established clinical tool used for the diagnosis and management of a broad range of diseases. With applications ranging from fetal imaging to echocardiography, its diagnostic use is exceeded only by two-dimensional

X-ray imaging [1]. US imaging uses sound waves above 20 kHz to construct images based on the interaction of the sound waves with the surrounding environment. The US waves, generated by a transducer (for imaging typically in the 1–20 MHz range), will be reflected, scattered, or absorbed by tissue and boundaries between tissues, such as skin and bone. The timing and strength of the returned echoes can then be used to determine the tissue location (assuming the speed of sound is known) and scattering characteristics with high spatial and temporal resolution. US images can be collected at over 20 frames per second, with a spatial resolution approaching 200 microns for higher frequency transducers. For a detailed overview of the physics behind US, please refer to the book by Richard Cobbold [2] (Fundamentals of Biomedical Ultrasound).

In addition to high temporal and spatial resolution of image acquisition, the popularity of US has been driven by notable advantages over X-ray computed tomography or magnetic resonance imaging (MRI), such as its well-established safety profile, low cost, portability, and broad accessibility. These benefits tend to outweigh some of the biggest challenges with US such as operator dependence and poor soft tissue contrast. The contrast and anatomical detail in US images has not been considered favorable compared with other modalities (e.g. MRI). One technology that has significantly improved US image quality and expanded US applications worldwide is the development of bubble-based contrast agents.

The advent of microbubble (MB) contrast agents has stimulated innovative strategies for cancer detection, therapy, and post-therapy monitoring, especially in the past two decades [3–5]. The use of gas as a contrast agent for US was first noted over 50 years ago [6]. Since then, several generations of this concept evolved from using unstabilized carbon dioxide (which was rapidly dissipated in the blood) to shell-stabilized constructs of more hydrophobic gasses. Current clinical MB formulations comprised a hydrophobic gas core (such as perfluoropropane—C₃F₈ or sulfur hexafluoride—SF₆) stabilized by a lipid, polymer, or protein shell. Bubbles generate contrast because of their interactions with the US waves. When placed in an acoustic field, bubbles

oscillate in response to positive and negative pressure changes; these oscillations are distinct from the surrounding medium because of differences in the compressibility and density between the gas and blood and the viscoelastic properties of the shell, as described in the following [7,8]. MBs have United States Food and Drug Administration approval for enhancement of echocardiographs and detection of liver lesions [4] in adult and pediatric [9] populations and have been investigated widely in numerous other clinical diagnostic applications. An excellent review of clinical diagnostic use of MBs and their safety profile compared with other contrast media was recently published by Erlichman et al. [5]. MBs have also been examined in numerous therapeutic applications ranging from neuro-modulation to sensitization of tumors to radiation therapy, and several approaches are currently in clinical trials [7,10–12]. Although a detailed summary of the current state of the art with MBs is beyond the scope of this review, we offer some additional thoughts about this field in the second part of this review.

MBs have had a significant impact in the field; however, MBs have some inherent challenges which may limit their utility in applications such as molecular imaging and image-guided therapy. The development of MB applications has been constrained by a relatively large particle diameter ($\sim 1\text{--}8\ \mu\text{m}$), which confines the MB to the vasculature, limiting their targets to upregulated biomarkers only within the intravascular space [13–16]. This reduces the potential for MBs to target biomarkers outside of the endothelium. MBs are typically produced at lower concentrations ($\sim 10^6$ to 10^8 bubbles/ml), which reduce their longevity both in ambient conditions and *in vivo*. Increasing the concentration is feasible, but leads to significant attenuation and shadowing, thus may not be practical to implement [17]. For clinical use, MBs are produced via on-site ‘activation’, which ranges from the hydration of a freeze-dried powder (Lumason®) to mechanical agitation of sealed vials containing lipid solutions and gas (Definity®). This process yields highly heterogeneous bubble populations, which range from submicron to $\sim 2\text{--}10\text{-micron}$ bubble diameters. It is worth noting that, in some cases, the polydispersity of MBs could be an advantage because it allows the use of many different frequency transducers and enables imaging of many anatomical targets. However, the general recent community consensus is that monodisperse bubbles could provide increased sensitivity and specificity to detect lower MB concentrations, critical for molecular imaging applications. Likewise, a reduction in bubble diameter to the nanoscale can complement MB capabilities and provide a robust platform for expanding the biomedical applications of US. Thus, in this review, we will focus on work associated with stable, echogenic, uniform NBs.

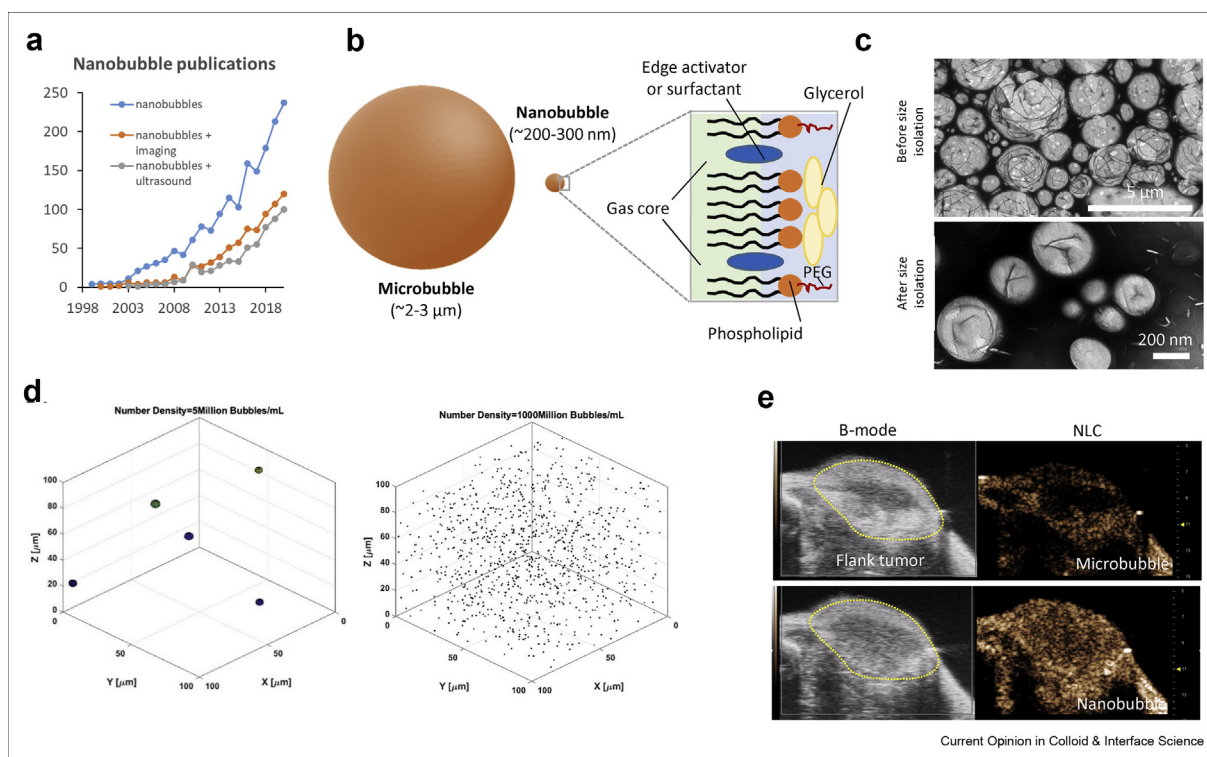
Nanobubbles (also frequently referred to as sub-micron or nanoscale bubbles, abbreviated here as NBs) have

been known to exist in commercially available MB formulations such as Definity® for nearly 20 years [18,19]. As mentioned previously, the MB formation, or “activation”, occurs via self-assembly of solubilized amphiphiles around a hydrophobic gas core. This relatively uncontrolled process results in a highly heterogeneous bubble population with a significant submicron component. However, as explained in detail in the following, the conventional theory would dictate that with decreasing bubble radius the backscatter from coated bubbles decreases significantly, and the resonant frequency increases rapidly. Thus, the submicron component of the acoustic activity was thought to be insignificant compared with that of the larger MBs ($\sim 1\text{--}5\ \mu\text{m}$ in diameter, which, by coincidence, has a resonant frequency close to the operating range of many commercial US imaging devices, about $3\text{--}5\ \text{MHz}$) and was largely dismissed. It was not until the intentional formulation of echogenic NBs by Oeffinger and Wheatley *et al* in 2004 and the first demonstration of their activity *in vivo* in 2006 by the same group [20,21] that the idea of NBs and their application as contrast agents in biomedical US became recognized. Concurrently, reports of circulating NBs coalescing in extravascular space into imageable MBs were published by Rapoport *et al*. [22]. However, as a field, echogenic NBs did not see rapid adoption until approximately 2010, when a burst of publications reported on their use in cancer imaging [23–31]. In the last decade, research into the application of NBs in diagnostic US and US-mediated drug delivery has seen rapid growth as shown in Figure 1A. The growth can also be correlated with the more widespread adoption of bulk NBs by the physics and colloid communities [32].

Early experimental reports of echogenic NBs garnered little attention from the contrast-enhanced US (CEUS) and/or MB and biophysics communities because of a lack of concrete, overwhelming evidence demonstrating that the measured or visualized acoustic activity was indeed stemming from the submicron bubbles and not from larger (near $1\ \mu\text{m}$ or above) bubbles which may have been contaminating the total bubble population.

Theoretically, the backscatter generated from the interaction of US with a particle is highly dependent on radius, with a $1\ \mu\text{m}$ particle scattering 10^6 times more than a $0.1\ \mu\text{m}$ particle at the clinical frequency range ($3\text{--}15\ \text{MHz}$, ignoring potential resonance). However, this assumes Rayleigh scattering independent of the effect of shell composition of the bubble oscillator such as shell viscosity, shell elasticity, and surface tension, as well as the acoustic pressure. At higher pressures and/or reduced surface tension of the bubble shell, the nonlinear activity of coated bubbles significantly contributes to its scattering strength. As described in detail later in this review, the current hypothesis is that the nonlinear activity due to the lipid shell coupled with the high particle

Figure 1



Overview showing (a) the rapid growth in publications from 1999 to 2020 focused in general on NBs (blue line), NBs in ultrasound-oriented applications (gray line) with the first publication in 2003, and more general imaging applications (orange line) with the first publication in the year 2000. (b) To scale representation of a microbubble and NB and the typical composition of a bubble used in ultrasound imaging. (c) Transmission electron microscopy (TEM) images showing the structure of bubbles immediately after self-assembly and after size isolation via differential centrifugation. The apparent heterogeneous mixture of bubbles prior is reduced to a fairly uniform NB fraction after the centrifugation process. (d) Simulation illustrating the striking difference in particle density per voxel of microbubbles compared with NBs for typical concentrations. It is likely that the high concentration strongly contributes to the observed NB acoustic activity. (e) Representative images of the same murine flank tumor showing microbubble (MicroMarker®) and NB enhancement in B-mode and nonlinear contrast mode. Images were acquired with the VisualSonics Vevo 3100 scanner at 18 MHz and 4% power. We thank Dr. Al De Leon, Dr. Eric Abenojar, and Mr. Hossein Haghi for their contributions to this figure.

concentration per imaging voxel is thought to be the major driving force behind the acoustic activity of NBs.

Recent progress in research using NBs for medical US imaging and drug delivery applications has been aided by the rapid developments in theoretical physics and experimental biophysics which provided strong evidence for the existence of bulk NBs as a viable, long-lasting construct [32]. Concurrently, several key experimental developments in imaging applications provided evidence that acoustically active NBs may be a viable contrast agent for US imaging. These were (1) the development of NB formulation methodologies with rigorous size control to reduce the size of the NBs to 200–400 nm and to remove all bubbles larger than 1 micron, which could be contaminating the solution and reducing control of acoustic response [28,30,33], (2) availability of instrumentation for quantitative, precise measurement of particle diameter, concentration, and, most importantly, buoyancy and total gas volume [34–36], (3) manipulation

of the NB shell to reduce interfacial tension and increase deformability using surfactants such as Tween 80 and Pluronic and edge activators such as propylene glycol, (4) the theoretical modeling of bubble acoustic response at lower surface tension and higher acoustic pressures [37,38], and (5) rigorously controlled *in vitro* and *in vivo* demonstrations of NB acoustic activity, extravasation, and imaging characteristics compared with MBs [29,39–41]. In the following sections, we detail these developments in the NB formulation and characterization. We then briefly discuss a theoretical framework for understanding the acoustic activity of NBs, provide an overview of the *in vivo* imaging and therapeutic applications, and finally discuss emerging future directions in this exciting, still-developing field.

NB formulation and characterization

NB formulation

A range of formulation strategies for gas-core nanoparticles has been explored. Many of the concepts rely

heavily on established processes from the fields of interfacial and colloid science and nanomedicine, where formulation of submicron particles, especially via self-assembly, has been thoroughly developed and vetted [42,43]. It is important to note that many different echogenic nanoparticles have been previously formulated, including echogenic liposomes [44,45] and nanodroplets [46]. However, this review will focus only on self-assembled gas-core, lipid, and surfactant-stabilized NBs.

Production methods of echogenic nanoscale gas bubbles use surfactants and additives to modulate the phospholipid membrane's viscoelastic properties and reduce the bubble's size to as low as 120 nm in both ambient and physiological conditions [23,47–52]. Most of the NB formulation methods evolved from MB preparation techniques. The primary mode of formulation is via simple self-assembly of various phospholipids (either a cocktail or a single lipid) or proteins and surfactants at the interface of an aqueous dissolution medium (such as phosphate-buffered saline, normal saline, or water). Lipids commonly used include 1,2-dipalmitoyl-sn-glycero-3-phosphocholine, 1,2-dibehenoyl-sn-glycero-3-phosphocholine, 1,2 dipalmitoyl-sn-glycero-3-phosphate, 1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine, and 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (ammonium salt). High lipid concentrations on the order of 10 mg/mL have been used to form the most effective, stable NB populations. This higher starting lipid content (which is $\sim 10\times$ higher than that for clinical MBs) is likely to contribute to high NB yield and a reduction in bubble coalescence [53]. Within this method are three distinct lipid dissolution processes—(a) shell materials dissolved in chloroform or another organic solvent before hydration of the lipid cake, (b) direct dissolution of all shell components in the aqueous medium, and (c) direct dissolution of shell components in propylene glycol before suspension in the aqueous medium [33]. Sonication and/or elevated temperatures can be used in each case to improve lipid dissolution. These processes can be performed with high volumes, increasing the formulation efficiency. The desired hydrophobic gas is forced into solution by mechanical agitation either using a modified dental amalgamator or a probe sonicator. This can be carried out in the bulk solutions or in individual sealed vials. After activation, the bubble solution is heterogeneous, and an additional step must be performed to isolate the NBs from this solution. More information is provided on this in the following.

Some notable exceptions to lipid NBs have been reported. Among these are (1) echogenic gas vesicles developed by Shapiro et al. [54,55], (2) poly (D,L-lactic-co-glycolic acid) (PLGA)-shelled NBs [25,31], (3) nested NBs developed recently [56], and (4)

nanocups developed by Kwan et al. [57]. Each of these constructs offers unique advantages and possesses distinct mechanisms of signal generation in an acoustic field. Gas vesicles are naturally occurring and can be genetically modified and remain stable for an extended period. The unique biologic origin enables extensive and predictable manipulation of these agents. PLGA-shelled NBs also offer greater stability than lipid-shelled bubbles. They are also easier to manipulate after formulation and have a significantly increased cargo capacity for drug delivery. However, both the gas vesicles and the PLGA-based NBs currently require high acoustic pressures for generation of detectable backscatter at clinically relevant frequencies. This also makes their echogenicity short lived. Additional development is needed to address these challenges. Although they are not fully coated gas bubbles, nanocups offer an interesting alternative formulation. These constructs are formulated by seeded polymerization of polystyrene coated with cross-linked polymethyl methacrylate. On dehydration and rehydration, the nanocups deform and create a cavity which entraps gas. These gas NBs, in turn, serve as cavitation nuclei on exposure to US pressures above 500 kPa and have been used in drug delivery applications. Nested NBs, developed recently, are lipid shell–stabilized gas bubbles of C_4F_{10} which are nested inside of liposomes. This formulation may stabilize the gas for an extended period, and is also a promising theranostic agent, because of the increased cargo capacity of the liposome.

On the importance of NB size isolation

Effective size isolation of NBs is essential for producing reliable and repeatable bubble formulations with well-defined diameters and for carrying out reproducible experiments, the results of which can be used to support the notion that the observed activity is derived from NBs, not a small population of MBs. Small numbers of MBs can significantly contribute to the overall US signal from a mixed NB solution. Hypothetically, for a NB concentration of 1×10^{11} bubbles/ml, even if 99.9% of the particles were in the submicron range, this would leave 0.1% (or 10 million) bubbles which were larger than 1 micron and could significantly contribute to the US backscatter. In some of the first published reports on applications of NBs in imaging, where the reported NB diameter was ~ 400 – 600 nm, it is possible that a significant portion of the signal visualized in *in vitro* and *in vivo* imaging applications was from MBs.

Techniques for improved isolation of NBs from the heterogeneous initial suspension include isolation by differential centrifugation [35,58] coupled with filtration by rigorously characterized syringe filters [59]. Both techniques, if implemented rigorously, significantly reduce MB contamination in the final suspensions. Because of the greater buoyancy of MBs, if left undisturbed, the solutions will separate into two

fractions, with the less or neutrally buoyant NBs remaining at the bottom portion within 1–2 h. However, to expedite the separation, differential centrifugation is typically used to effectively isolate NBs and remove most, if not all, MBs from the solutions. When centrifuged at 50 g for 5 min, all bubbles larger than 0.7 μm should rise 0.5 cm or more. As an additional or as a standalone step, filtration also can be used. Here, either a sequential or single filtration process with well-characterized syringe filters can reduce the NB diameter and further reduce the dispersity of the population [59]. Overall, NBs formulated using this process can be made at a high concentration ($10^{11}/\text{ml}$) with diameters ranging between 100 and 600 nm, are highly reproducible and scalable, and tend to be significantly more stable *in vivo* than clinical MB agents [35,60,61]. However, characterization of NBs remains a challenge despite many promising technologies.

NB characterization

Several standard techniques are used to quantify nanoparticle size distribution and concentration. These include dynamic light scattering (DLS), transmission electron microscopy, cryo-electron microscopy [36], nanoparticle tracking analysis [62], and the Coulter counter [63]. Each of these techniques comes with caveats, and when used alone, none provide an accurate representation of the nanoparticle size distribution and concentration. Thus, multiple methods should be used to validate any NB formulation. For diameter measurement, DLS is high throughput and is considered the industry standard for measuring the nanoparticle population's size distribution. The technique is highly sensitive and can measure a broad size range from several nanometers to >10 microns and is independent of particle density. Zeta potential measurements are also conducted using the DLS equipment. However, DLS does not measure particle concentration. To measure nanoparticle concentration in addition to size, instruments such as the Coulter counter, nanoparticle tracking analysis (NTA), and resonant mass measurement (RMM) are available. Because formulation via self-assembly is not well controlled, the resulting particles are highly heterogeneous, spanning <10 nm to >5 microns. In unprocessed solutions, the mean diameter is roughly 1–3 microns. These bubbles are typically characterized via the Coulter counter because this technique provides information about the particle size and concentration. However, as reported previously, the Coulter technique has some challenges with detection limits depending on the aperture used, as well as with coincidence artifacts (where several particles are counted as a single one) [64]. The Coulter counter also cannot determine whether a particle is filled with gas or a solid undissolved lipid out of solution or a micelle or a giant unilamellar vesicle or dust. The detection limit for

larger apertures tends to hover around 1 micron to 500 nm and at the smallest aperture as small as 200 nm.

RMM is the only method capable of distinguishing particle buoyancy. The technique has been validated and well established as a useful tool for NB characterization [34,35]. Experiments have shown that even with well-developed formulation protocols, significant nonbuoyant populations of nanoparticles are present in solutions. Although these nonbuoyant populations of nanoparticles do not have significant acoustic activity, the particles will be counted in techniques such as the Coulter Counter or nanoparticle tracking analysis, which can potentially overestimate the actual gas-core bubble concentration in a formulation. However, RMM is not without caveats. First, it relies on a microfluidic sensor, limiting the size of the nanoparticles passing through it. The nanosensor enables measurement of up to 2 microns. To measure larger particles, the microsensor must be used. However, this sensor cannot detect smaller particles. As such, both sensors should be used for each heterogeneous solution, making the technique somewhat cumbersome. For buoyant particles, the limit of detection tends to be near 100 nm. Thus, many smaller micelles in the solution will not be counted. Finally, the particle diameter calculation relies on input parameters such as the particle density, which the user provides and can be difficult to establish for new formulations.

Ultimately, at least two techniques should be used to characterize a NB formulation. These should be capable of measuring diameter with a broad limit of detection (DLS) and accurately determining the particle concentration (Coulter or RMM). Proper technique development and validation are essential yet quite difficult. A rigorous, quantitative measurement of the gas contained within a bubble can be used to validate any of the abovementioned measurement methods. Headspace gas chromatography/mass spectrometry is a well-established, highly quantitative technique that can be used to validate measurements based on Coulter, RMM, NTS, and to some degree DLS.

Manipulation of surface tension and shell viscoelastic properties

A reduction in the interfacial tension of the stabilizing shell reduces the Laplace pressure of a gas bubble. Several surfactants have been explored for this application in the NB community. The most prevalent of these is the nonionic surfactant Pluronic (also known as poloxamer). Various Pluronics have been explored as NB additives, with the most frequently used being Pluronic L10, F68, and L61. The effect of varying the Pluronic composition has been extensively investigated. Furthermore, shell deformability and buckling have also been shown to contribute significantly to the acoustic bubble response and bubble stability both within the

acoustic field and under ambient conditions. Here, edge activators, such as propylene glycol, which have been used commonly to increase liposomes' deformability and increase penetration through layers of the skin and liposome literature, have been investigated [65]. Propylene glycol also acts as a solvent for lipids and can be used to eliminate chloroform in NB formulation [33].

To summarize, the formulation of NBs yields particles of <100–600 nm in diameter and a high particle concentration (on the order of 10^{11} bubbles/mL). The resulting NB features were similar to conventional nanoparticles, such as micelles. At the same time, several characteristics distinguish NBs from other nanoparticles. Foremost is the lower particle density due to a gas core. Lower density particles have been shown to marginate more effectively in flow, increasing the probability of extravasation in hyperpermeable vasculature [66]. Another feature is a highly deformable shell combined with a compressible gas core that enables better penetration and extravasation [67]. Nanoparticle density, deformability, and concentration [68] have all been shown to improve tumor accumulation. Consistent with these findings, intact, acoustically active NBs have been shown to readily extravasate and accumulate in tumors [39,69,70]. These features make NBs useful in various applications described in the following.

Theoretical considerations and simulations

What can explain the contrast seen in the published work? It is not expected that NBs would significantly contribute to US backscatter at conventional clinical imaging frequencies. The wavelength of US at clinically relevant frequencies (1–20 MHz) is approximately 1500–75 μm , which is orders of magnitude greater than the submicron NB size ($\sim 0.2 \mu\text{m}$). Moreover, the resonant frequency of an uncoated bubble can be calculated from [71].

$$f_r = \frac{1}{2\pi R_0} \sqrt{\frac{3\gamma P_0}{\rho} \left(1 + \frac{2\sigma}{P_0 R_0}\right) - \frac{2\sigma}{\rho R_0}}$$

where R_0 is the radius of the bubble, P_0 is the ambient pressure, γ is the polytropic coefficient, ρ is the density of the medium (water), and σ is the surface tension of the medium. For a 0.2 μm (200 nm) air bubble, the resonant frequency is ~ 115 MHz (the addition of a bubble shell increases the resonant frequency). Therefore, NBs are far from their resonant frequency when exposed to 1–20 MHz US waves.

More generally, it is not expected that NBs have a large scattering cross section, even if the compressibility and density contrast is large (gas/fluid). Solutions to the Anderson model of a fluid sphere (a gas can be considered a liquid) reduce to that of a Rayleigh scatterer

when the wavelength of the US is much greater than the size (a) of the bubble:

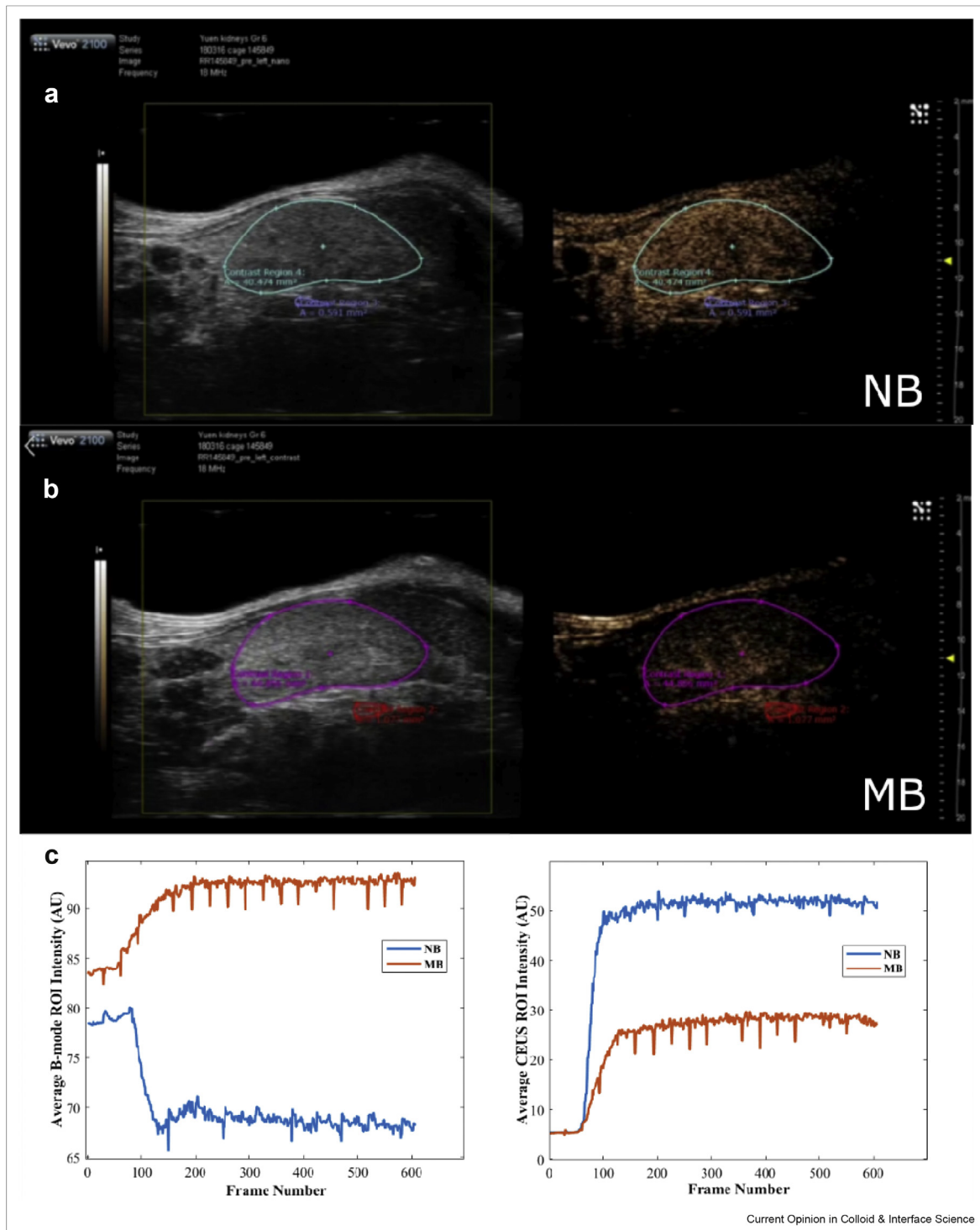
$$\sigma = 4\pi a^2 (ka)^4 \left[\left(\frac{K - K_0}{3K} \right) + \frac{1}{3} \left(\frac{(\rho - \rho_0)}{(2\rho + \rho_0)} \right)^2 \right]$$

Because the scattering cross section is proportional to a^6 , it is expected that scattering from a $a = 1 \mu\text{m}$ MB is 10^6 greater than a 0.1 μm (200 nm diameter) bubble (ignoring that the MB may be close to resonance depending on the excitation frequency). However, the volume differential of a MB compared with a NB is proportional to r^3 . For a MB that is 10 times greater than the NB radius, the gas volume is 10^3 greater. Therefore, it would be difficult to simply increase the concentration of NBs to compensate for their significantly reduced scattering compared with MBs. If NBs can achieve similar echogenicity to MBs for clinically relevant US frequencies, there must be other factors that contribute to their enhanced echogenicity.

A critical insight was provided in our first experiments comparing MB and NB contrast in living mice. The experimental data presented in Figure 2 (unpublished data) compare the injection of commercial MBs (Vevo MicroMarker, FUJIFILM VisualSonics, average size 1.8 μm) [72] with custom-made NBs (average size 263 nm) [33]. The injection volumes (5 ml phosphate-buffered saline + 50 ml bubble solution) were comparable for both agents. The concentration of the NBs was 1×10^{10} NB/ml and for the MBs, 2×10^8 MB/ml. CEUS images were acquired with a Vevo 2100 (FUJIFILM VisualSonics) at 1 fps at 18 MHz. Simultaneous B-mode and nonlinear contrast-enhanced images were acquired. This allows a direct comparison of (a) changes in US backscatter and (b) changes in the nonlinear scattering before and after the injection of the NBs. The video (Figure 2 and S1) shows that the NBs have comparable contrast with the MBs, but there are apparent differences in the time–intensity curves (TICs) and the spatial dependence of the kidney TICs. The CEUS images are based on an amplitude modulation (AM) method [73]. Consecutive US pulses of identical shape but with different amplitudes (e.g. a factor of 2) result in similar reflections from the linear scatterers but with an amplitude difference (x2). When scattered by the nonlinear NBs, the same two pulses would differ in both the amplitude and shape because of the bubble nonlinear response. For *in vivo* imaging, this would suppress a significant portion of tissue scattering, which is linear, and amplify bubble scattering, which is nonlinear. This is seen in Figure 2.

Supplementary video related to this article can be found at doi:10.1016/j.cocis.2021.101463

Figure 2



Comparison of MB and NB contrast kinetics. Ultrasound B-mode images (left) and CEUS ultrasound imaging (right) of the mouse kidney in the last frame collected (data collected at 1fps) for (a) NB injection and (b) MB injection. In (c), the time kinetic curves for the mean intensity in the selected ROIs for B-mode (left) and CEUS (right) for the NB injection (blue curve) and MB injection (red curve). Video(s) of the above mentioned injections can be found in the Appendix.

The B-mode and CEUS images using MBs are presented in the bottom row. The same experiment was conducted with NBs approximately 25 min later, in the same mouse. The data are shown in the top row. For the NB data (top row), it can be observed that for this concentration, there is a significant enhancement in CEUS images. Although there is a significant increase in the nonlinear CEUS signal, there is a slight decrease in US backscatter with the introduction of the NBs (as assessed by the average intensity in the region of interest (ROI) of the B-mode images). If the introduction of the NBs was to increase the US scattering overall, this would have been detected as an increased intensity in the B-mode image. Some of the differences between the MB and NB formulations can be attributed to the increased attenuation of the NB formulation (because of the higher concentration, evident from tissues under the kidney) and/or acquiring the images in different kidney imaging planes (even though an effort was made to image the same plane). Nevertheless, the difference in the ROI data in Figure 2 supports the hypothesis that the key to the CEUS signal of the NBs is not their intrinsic echogenicity, but their nonlinear response to US.

When using a bolus injection of MBs in the same kidney, there is also a significant change in the CEUS images. However, in this case, the kidney filling dynamics are different: contrast enhancement is first apparent in the renal and interlobular arteries of the kidney, followed by cortical enhancement. Contrast enhancement is predominantly localized to the kidney with a minimal signal in the surrounding tissues. Moreover, there is an increase in the intensity in the B-mode images, indicating that contrast is enhanced without the addition of specialized CEUS sequences. An increase in the B-mode image intensity is not observed during the NB injection.

We have hypothesized that the key to the lipid shell NB effectiveness as a contrast agent is the nonlinear shell property. It is important to distinguish two types of nonlinear bubble oscillations: (a) inherent nonlinear oscillations of the bubble oscillator (in which shell properties are linear) and (b) the enhanced nonlinearity of a lipid-coated bubble (in which the shell properties are nonlinear). This enhanced nonlinearity produces rich harmonics, even at very low pressures (as low as a few kPa), making NBs very effective in CEUS modes.

The nonlinear shell properties can contribute to the enhanced echogenicity in two ways: (a) decreasing the effective resonance frequency of the NB and (b) providing significant contrast due to nonlinear oscillations and a sharp pressure-dependent bubble response. It has been shown that increasing the US pressure incident on an MB decreases their resonant frequency, a phenomenon known as pressure-dependent resonance

[74]. The pressure-dependent resonance is more significant for lipid-shelled bubbles, with a strong dependence on the initial surface tension on the NB⁵ (and the shape of the curve that describes the surface tension curve as a function of the bubble radius). For example, in coated bubbles with an initial surface tension $\sigma_0 = 0.01$ or 0.062 N/m, the pressure-dependent resonance frequency can occur at frequencies as low as half the linear resonant frequency. Therefore, we expect that the nonlinear shell properties significantly contribute to enhanced echogenicity at lower frequencies. However, this effect would not explain the lack of increased NB echogenicity in the B-mode imaging in Figure 2, while observing at the same time high contrast-to-tissue ratio (CTR) in the CEUS images. In these images, the strong contrast is likely due to the use of the AM CEUS techniques. This is likely because rich harmonics are predicted from the NB oscillations [74–76], providing strong contrast compared with tissue. Moreover, the properties of the lipid shell enhance threshold-dependent nonlinear oscillatory behavior. Saddle node bifurcations in the analysis of bifurcation diagrams, as shown by Sojahrood et al. [75,77], result in the abrupt enhancement of the scattered US above a specific pressure threshold, substantially increasing contrast in amplitude-modulated CEUS schemes. As the CEUS scheme implemented on the commercial instrument is based on an AM technique, a significant contributor to the enhanced echogenicity is likely the pressure-dependent threshold for the sudden increase in NB echogenicity. Therefore, it is postulated that the key to understanding the unexpected echogenicity of lipid-coated NBs is the highly nonlinear response of the bubble oscillator.

Importance of informed optimized imaging hardware and software

There is some evidence that CEUS contrast using NB contrast agents can be achieved at conventional US frequencies of using standard onboard nonlinear imaging pulse sequences (using transmit frequencies ranging from 3 to 9 MHz) [48,70]. Moreover, the highly nonlinear NB response to US can be achieved at low pressures. Therefore, in principle, the only specialized US hardware required to achieve robust contrast is the existence of imaging modes that rely on the efficient detection of nonlinear signals (harmonics, subharmonics, superharmonics) and especially modes that rely on processing subtraction algorithms (such as pulse inversion [PI]/AM schemes) that can suppress background tissue echoes. Subtraction algorithms have significant advantages at higher frequencies. The energy of the nonlinear tissue signal at the higher harmonics (e.g. second) is more dominant at high frequencies than that at low frequencies, reducing the CTR for harmonic CEUS imaging. Moreover, because US attenuation increases with frequency in soft tissues, the higher harmonic components at high frequencies will undergo higher attenuation

than at lower frequencies. Therefore, optimized US parameters for CEUS imaging will be highly dependent on the application [72]. Finally, the TICs for NBs have different kinetics than MBs since NBs. Because of the NB size, the NBs potentially extravasate into tissues. Therefore, the contrast kinetic distribution (which depends on factors such as the blood flow and volume, vessel density, and vessel permeability) will be different [78,79]. This may necessitate different approaches for the interpretation of semiquantitative parameters in the TIC (e.g. area under the curve, time to peak, wash-in time, wash-in rate, wash-out time, wash-out rate), including more sophisticated compartmental modeling to describe the NB transport in tissues [78].

Monodisperse NBs

Other investigators have shown that monodisperse MBs have a more uniform acoustic response and increased imaging sensitivity than polydisperse MBs [80,81]. Consequently, this may also apply to monodisperse NBs when compared with polydisperse NBs. Using microfluidics techniques we have previously reported on [82,83], we have developed a new approach to make monodisperse NBs. The method uses a two-component gas mixture of water-soluble gas (nitrogen) and water-insoluble gas (octafluoropropane) to first generate monodisperse MBs with a microfluidic flow-focusing junction. The MBs shrink because of the dissolution of the water-soluble components in the gas mixture. The degree of bubble shrinkage may be precisely controlled by tuning the ratio of water-soluble to water-insoluble gas components. This technique maintains the monodispersity of the NBs and in recent experiments has shown similar advantages to those of monodisperse MBs [84].

Imaging applications of NBs

Long-circulating contrast agents that can cross leaky tumor vasculature and penetrate into tissue provide a clear opportunity to augment the applications of CEUS. Many diseases, including type 1 diabetes and cancer, have hyperpermeable vasculature [85]. The field of nanomedicine has taken advantage of, and in some cases hotly debated, the role of vascular permeability in the delivery of therapeutics, especially in cancer. Overall, however, it is generally accepted that vascular permeability does increase tumor accumulation of nanoparticles [86]. Building on this concept, NBs have also been able to exploit enhanced vascular permeability to image tumors. Extravasation of the NBs to target biomarkers expressed in the tumor microenvironment and on cancer cells makes the detection and delineation of tumors more effective and efficient. The small size, highly deformable shell and low density facilitate rapid NB margination and extravasation through leaky vasculature. Furthermore, NBs can be made at a high concentration (10^{11} /ml). This increases scatterer number

density, improving the sensitivity of detection and likelihood of extravasation into tumors, as demonstrated recently [68]. Evidence supporting the extravasation of NBs into tumors and the diabetic pancreas in mice is shown in Figure 3.

Consideration of imaging parameters

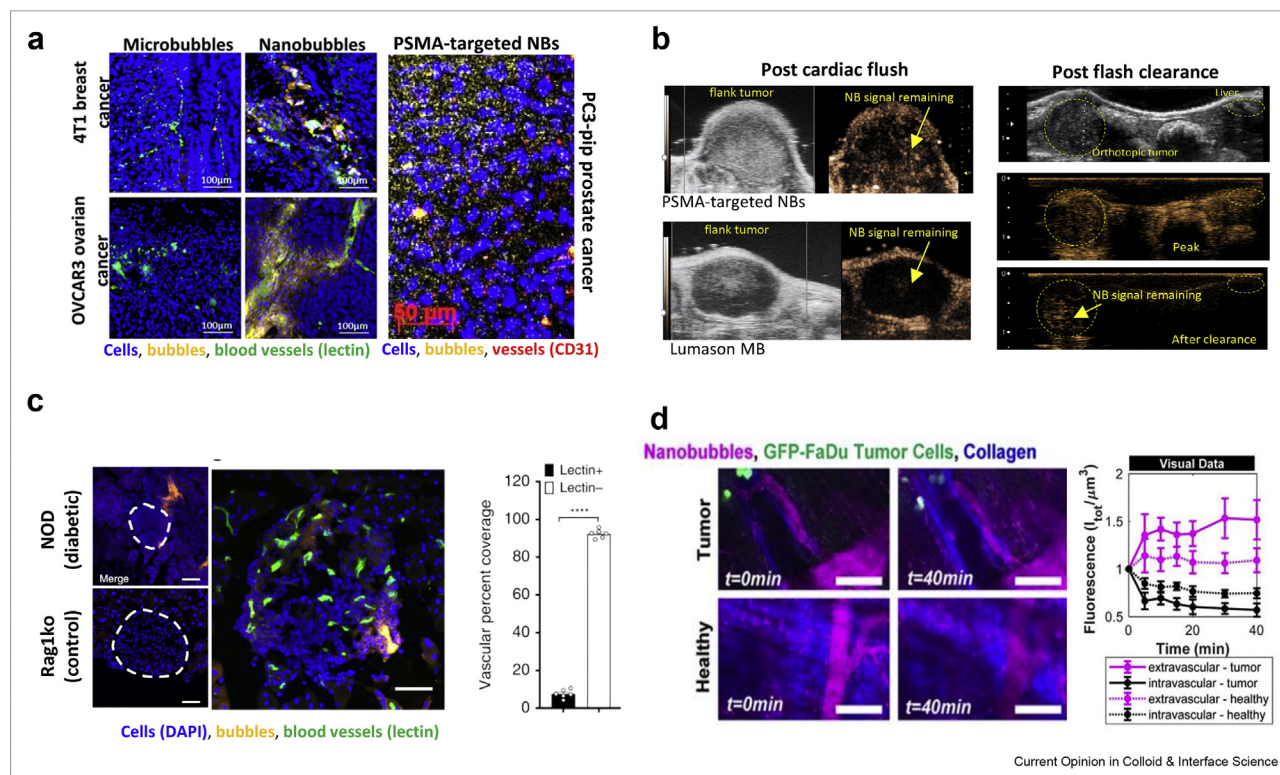
As mentioned previously, bubbles scatter a large component of the incident pulse energy nonlinearly compared with tissue. Nonetheless, to achieve high sensitivity to bubble echoes *in vivo*, tissue backscatter must be removed while amplifying signals from the contrast agent. To accomplish this, most clinical scanners use a combination of PI and AM to maximize the CTR [87]. PI uses two US pulses sent consecutively into the tissue, with the second pulse being an inverted copy of the first and summing the resulting echo signals. This removes linear scattering. AM detects ultrasound contrast agent (UCA) echoes by sending two US pulses into tissue, scaling the signal in response to the second pulse, and subtracting successive echo signals [87]. This further removes linear scattering. For NBs, which respond better at higher frequencies, there are two confounding issues with the PI and AM methods. First, the amount of the nonlinear tissue signal is greater at high frequencies than that at low frequencies [88]. Second, US attenuation increases as a function of frequency in soft tissues [2]. For these reasons, optimizing the frequencies and amplitudes of the transmit pulses to maximize the CTR specifically for NBs, compared with existing MBs, is a significant unmet need in the field.

The use of parametric dynamic CEUS (DCE-US) imaging has been explored in numerous studies to measure contrast kinetics and assess tissue perfusion clinically and preclinically [89–93]. The technique relies on MB contrast agents and exploits the real-time image acquisition and high signal-to-noise ratio of US to measure MB kinetics. Using nondestructive low mechanical index (0.06–0.2 depending on the vendor) nonlinear imaging, DCE-US is used to track the change in the signal in a region of interest over time (represented by the TIC) on a per pixel basis; these curves can be fitted with mathematical models related to tissue perfusion. DCE-US has been used to assess tumor angiogenesis and antiangiogenic therapy response [89,90,94,95]. Because MBs are exclusively a blood-pool agent, DCE-US may enable accurate perfusion quantification. The NB signal could also provide kinetic information about tumor vascular permeability. Another unmet need is developing new pharmacokinetic models suitable for NBs and targeting NBs, which reflect extravascular activity in tumors [96].

Targeted NBs in imaging applications

A variety of NB constructs have been developed and tested in *in vivo* imaging applications. The platform nanoparticles in these studies have typically consisted of

Figure 3



NB extravasation quantified in a variety of models. (a) Immunohistochemical analysis of fluorescent NB distribution in 4T1 breast and OVCAR 3 ovarian tumors after bubble injection IV via the tail vein. (b) Extravasation of acoustically active NBs assessed via ultrasound in animals after cardiac flush and in living mice; in the latter case, circulating bubbles were destroyed peripherally via repeated high-intensity flashes using a clinical transducer. (c) Representative confocal images of an islet within a pancreas section of a 10-week-old female nonobese diabetic (NOD) and 10-week-old Rag1ko control mouse after rhodamine-labeled NB infusion (orange). The right image shows the maximum-projection confocal image of an islet of the NOD mouse and the corresponding mean NB coverage in nonvascular areas (lectin negative) and vascular areas (lectin positive). (d) Intravital microscopy in conjunction with simultaneous acoustic emission measurement in a window chamber model. In all cases, the extravascular echogenic NB presence is clearly visible, providing strong evidence that NBs indeed can extravasate and accumulate in the extravascular tissue. Data also support that the extravasated NBs are able to retain, at least in part, their acoustic activity. (a) Adapted with permission from Wu *et al* *Ultrasound in Medicine & Biology*, 45(9), 2019 Figure 7 and Perera *et al*, *Nanomedicine NBM*, Volume 20, 2020, Figure 5; (b) Unpublished data; (c) Adapted with permission from Ramirez *et al* in *Nat Commun* 11, 2238 (2020). <https://doi.org/10.1038/s41467-020-15957-8>, Figure 2; (d) Adapted with permission from *Theranostics* 2020; 10(25):11, 690–11706. doi:10.7150/thno.51316, Figure 6.

a lipid shell with additives, including porphyrins [97] and/or fluorescent dyes used for concurrent optical imaging and immunohistochemical analysis [98]. The main modifications and innovations in most published reports focus on surface functionalization approaches to create targeted NBs. Targeted NBs are attractive agents for molecular imaging and can detect biomarkers on target cells within the vasculature and beyond. Notably, the active targeting moieties have been shown to increase bubble extravasation and retention in tumors compared with untargeted NBs [69]. This approach is attractive for increasing the sensitivity of detection and improving drug delivery efficiency. A range of ligands have been explored. Ligands can be conjugated to lipids via standard conjugation chemistry, including the EDC/NHS (1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide/N-hydroxysuccinimide) reaction mechanism and the use of

maleimide–thiol coupling chemistry. Early work also included the use of avidin/biotin complexes to functionalize the NBs, but more translational approaches have mostly replaced this approach. Full antibodies [99], peptides [98], affibodies [100], and nanobodies [101] have all been used as targeting moieties.

Two of the most developed areas for targeted-NB applications are prostate cancer and breast cancer imaging and therapy. These areas have a potentially significant and more immediate clinical impact because the use of US is common in the clinical workflow for disease diagnosis. In prostate cancer, the use of transrectal US is nearly ubiquitous during biopsies. Here, US is used to localize the prostate gland [102]. In breast cancer, US is used commonly to complement mammography and to guide biopsies of suspicious lesions. Thus, innovations

in US contrast agents can increase cancer detection sensitivity and specificity with minor changes to the existing process. Targeting the prostate-specific membrane antigen in prostate cancer via various NBs has been investigated [69,101,103,104]. In breast cancer, using Herceptin to target the Her-2 receptor and folate to target the folate receptor has shown promising results [105,106].

Additional noncancer applications have been investigated using NB CEUS. For example, NBs were applied to detect asymptomatic type 1 diabetes in mice as recently reported by Ramirez et al. [107]. Because NBs can permeate the leaky vasculature, the NB signal within the endocrine diabetic pancreas was higher than that from normal mice. NBs were retained within the islets for an extended period. Furthermore, the agents showed increased preferential NB accumulation consistent with the severity of insulinitis, allowing potential disease progression assessment.

There are three key challenges in NB development for diagnostic imaging applications. The first is a rigorous and meaningful interpretation of the TIC data when using dynamic parametric CEUS. The change in signal intensity over time within a region of interest (typically within a single imaging plane) can be a valuable parameter for describing vascular architecture in tumors and examining vascular density and heterogeneity and their changes in tumor development or after tumor treatment. However, many factors can influence the contrast kinetics and confound interpretation. These include biological effects such as macrophage uptake and more general removal via the reticuloendothelial system and acoustic parameters such as the mechanical index and the imaging frame rate. The acoustic parameters determine the agent's total exposure to US energy and, therefore, the longevity of the contrast agent exposed to US excitation. Likewise, the natural rate of dissipation of the gas from the agent shell will also govern the *in vivo* circulation time.

The second challenge is that target-bound NBs are vulnerable to rapid decay because of the continued exposure to the acoustic field. This could, in turn, contribute to more rapid signal decay compared with circulating NBs. However, if properly validated, TIC parameters such as the area under the curve and rate of signal wash-out could be key indicators of these processes and have been shown to differ in targeted versus untargeted NBs [69]. Unraveling the extravascular and intravascular contributions to the TIC data is a key remaining challenge. For this reason, appropriate controls are needed for proper TIC interpretation. These controls should include MBs, which should describe the blood pool agent kinetics. However, caution should be taken to remove submicron bubbles from these solutions, which could complicate the MB

data interpretation. Some nonspecific, nonselective accumulation is expected because of passive extravasation and cell uptake of untargeted NBs and may confound the interpretation of imaging data. However, published studies comparing the acoustic activity of targeted versus untargeted NBs *in vivo* have shown that the washout of targeted NBs is significantly reduced compared with their untargeted counterparts [41]. This reduced washout may have to do with specific cellular internalization pathways, which may stabilize the targeted NBs once they have entered the target cell. If this is shown to be the case, then the nonspecific NB signal can be reduced given sufficient time (~ 30 min in mice), and the remaining signal is likely to be specific to the target.

Another challenge is the influence of bubble concentration and gas volume on bubble decay. Studies have shown that in both MBs and NBs, the signal's decay rate increases with a decreased concentration and corresponding gas volume. Thus, these factors must be accounted for between experiments and carefully compared. If the optimal diagnostic agent and imaging properties are to be investigated, these will vary between formulations and between NBs and MBs. Thus, what parameter should be normalized and how controls should be designed remain topics of investigation.

Therapeutic applications

Another key area of growth for NBs is in image-guided therapy. The use of NBs in image-guided therapy follows the well-developed concepts in nanomedicine. For example, a large body of work in nanomedicine has focused on exploiting the enhanced permeability and retention in tumors to deliver the therapeutic payload [108]. The same challenges that have held back the development of successful nanoparticle-based drug delivery systems in cancer chemotherapy apply to NBs. These include the same physiological barriers to efficient delivery—foremost nanoparticle–blood interactions, scavenging of the nanoparticles by the reticuloendothelial system, complex blood–nanoparticle interactions, limited margination in flow, extravasation via the leaky endothelium, and sufficient distribution within the tumor parenchyma. Another set of barriers, including the NB interaction with target cell membranes, cellular internalization, endosomal entrapment and escape, exist [86,109]. These processes are relevant to NBs' diagnostic/functional imaging and therapeutic potential and need to be studied in depth.

However, when applying the decades of experience from nanoparticle transport studies to NBs, it becomes apparent that NBs have key advantages that make them uniquely suited for overcoming some of these barriers. As mentioned previously, these are (a) buoyancy and low

density that facilitate margination in blood vessels, (b) high concentration, and (c) deformability owing to low elasticity, high deformability, and low interfacial tension of the shell. These factors increase the probability of decreased reticuloendothelial system (RES) uptake and increased extravasation and movement through the extracellular space. This makes NBs an exciting vehicle for drug delivery to tumors and other pathologies.

Furthermore, NB transport and NB drug delivery can be aided by US-driven cavitation and other US bio-effects—processes unique to this approach. The use of US for active, on-demand delivery of drugs is a promising area of investigation. In these approaches, acoustic waves lead to an increase in the local energy absorbed, acoustic fluid streaming, cavitation of gas bubbles, and mild hyperthermia, which, in turn, increase vascular permeability or trigger payload release from the carrier [110]. These strategies either are applied to various existing drug-loaded nanoparticles such as liposomes [111–114] or use US-visible MBs (1–10 μm in diameter, which remain in the blood vessels and cannot directly reach tumor cells) in combination with therapeutics [115,116]. It is important to note that, on injection into the bloodstream, the relatively low number density of MBs accessible to the target site at any one time is likely too low to elicit a strong therapeutic effect. In contrast, NB number density, which is 3 orders of magnitude greater, can provide sustained high levels of particles at the target site, which is likely to improve therapy outcomes. Figure 1D illustrates the difference between the two formulations.

Nanoparticles are not easily visible on US at clinically used frequencies without specialized imaging modes because of their small size and correspondingly high resonant frequency. In contrast, MBs generate strong contrast on clinical equipment, and their cavitation can induce considerable bioeffects of vasculature and cell

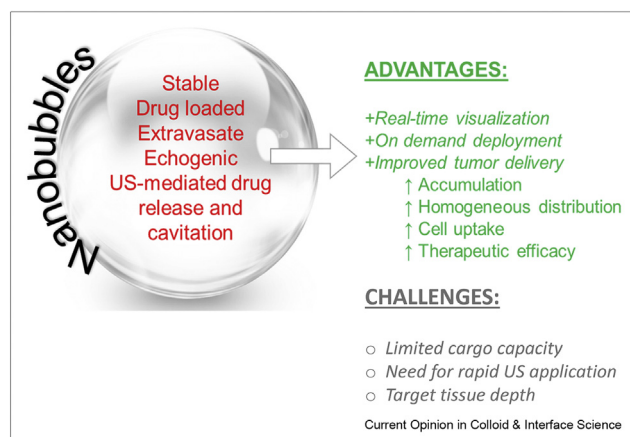
membranes [7]. However, they do not extravasate beyond the vasculature and are thus of limited utility in *in vivo* applications, which require them to be located proximal to the tumor cells. NBs combine characteristics of MBs and nanoparticle drug carriers into one simple platform that allows for a significant increase in the therapeutic index of the delivered agent by penetrating the physiological and cell barriers that limit MB accessibility to intended target sites [31,56,117]. The highlights are summarized in Figure 4.

Other unique therapeutic applications of nanoscale contrast agents move beyond US-mediated drug delivery. These techniques often rely on highly localized US-induced cavitation (both stable and inertial) of bubbles at the target site. For example, gas vesicles have been recently used as molecular reporters for US imaging of gene expression *in vivo* [118] and neuromodulation [119]. NBs have been used as sensitizing probes for tumor-specific radiation therapy of prostate cancer [120] and as robust agents for opening the blood–brain barrier [121]. Overall, the capacity to elicit precise therapeutic effects using US-sensitive nanoscale contrast agents, whether drug-loaded or not, promises to enhance disease management.

Future perspectives and best practices

Key innovations will drive the field forward. Foremost, the efficient and reproducible production of NBs with a narrow, or ideally monodisperse, size distribution and high yield (concentrations above 10^{11} NBs/mL) is critical to successful implementation of these agents. Developing monodisperse bubbles can amplify their acoustic activity and further increase the contrast to the background signal in the target tissue. Optimizing US exposure parameters to monodisperse NBs may limit clinical use to a narrow range of applications and decrease the robustness of the agents to natural human variability. Nevertheless, the large increase in sensitivity demonstrated with monodisperse MBs when compared with polydisperse MBs suggests that this trade-off may be warranted [122]. In combination with producing more monodisperse bubbles, shell engineering—manipulating the viscoelastic properties of bubble coating—will continue to yield significant benefits. Going a step further, model-based rational shell design can yield highly predictable, uniform, and tunable acoustic responses. Theoretical modeling and numerical simulations can provide unique insight into the shell effect on the observed controllable acoustic behavior. Both of these advancements can aid in the production of NB libraries of established sizes that can be individually distinguished solely on their resonant frequency shifts or pressure-dependent nonlinear behavior. Such developments will serve to enable multiplex or multicolor US imaging to distinguish two or more targets simultaneously in the same imaging plane. The innovation in

Figure 4



Summary of NB features and challenges in drug delivery.

this space would inspire significant advancements in the molecular imaging field.

With all new formulation modifications, rigorous characterization of NBs must be carried out to confirm the (a) concentration, size distribution, stability, and (b) the acoustic activity of the NBs under physiological conditions, especially in whole human blood. The techniques need to have high sensitivity and a broad working range to detect buoyant particles of smaller diameters than currently feasible with RMM. Currently, the best practice is to use a suite of instruments (DLS, Coulter, RMM, transmission electron microscopy, SEM) coupled with headspace gas chromatography–mass spectrometry validation. In addition, working in concert with bubble formulation, the optimization of imaging hardware (such as broader development of high frequency and 3D enabled transducers able to collect nonlinear data in real time with a high frame rate), acoustic imaging parameters, pulse sequences, and pharmacokinetic models to best take advantage of the unique NB properties will be necessary to advance the field. Characterization of acoustic activity in a bubble population and on a single bubble basis would also provide insights into the acoustic activity for each new agent. Assessment via passive cavitation detection over a range of pressures and frequencies is necessary, especially for therapeutic formulations. This has already been recognized as an important dosimetric consideration in MB therapy [123] and will be necessary to implement for NB applications.

Multifunctional NBs are another area that is likely to see significant interest. NBs, which are magnetic via the inclusion of iron oxide nanoparticles within the shell, have been shown to have dual functions as US and MRI contrast agents [124]. They are also being applied as therapeutic moieties. Likewise, combinations of a strong optical absorber and gas core have led to the creation of dual-mode, acoustic, and photoacoustic imaging agents, which have many exciting applications. Finally, multimodality, MR, photoacoustic (PA), and US imaging and optical/fluorescent imaging probes are under development and can enable successful imaging and therapy and biopsy/surgical resection guidance [98,125].

Inflammation is associated with a broad array of diseases, from cancer to osteoporosis, rheumatoid arthritis, central nervous system (CNS)-related diseases such as Alzheimer's disease, and cardiovascular pathologies. Therefore, in principle, NB CEUS applications can be expanded to aid in the diagnosis, monitoring of progression, and assessing treatment outcomes of a broad collection of pathologies linked to inflammation. In addition, treatment of these diseases via NB-enhanced or NB-delivered technologies offers considerable promise. In cancer specifically, NBs are poised to have a role in more effective early diagnosis of a variety of

tumors and early detection using novel biomarkers. Guidance of procedures—such as tumor biopsy and resection—is also an exciting area of investigation. Enhanced permeability and retention monitoring is another synergistic role for NBs that can be critical in potentially driving the field of nanomedicine forward. It may be possible that NB transport, as quantified by real-time contrast-enhanced imaging, can serve as a tool for assessing vascular permeability and the extent of endothelial fenestrations. Can we apply the knowledge in pharmacokinetic modeling, combined with super-resolution signal and image processing techniques, to quantify NB transport in a tissue of interest with high sensitivity? If these techniques can be adequately developed, NBs may then serve as a tool for assessing 'extravasation potential' of this target tissue and be used to predict therapeutic efficacy of nanomedicine-based therapeutics. Finally, the use of NBs to elicit an immune response due to mechanical disruption of cells *in vivo* (abscopal immune responses [126]) and monitor cancer immunotherapy outcomes via targeting of inflammation markers are also an exciting emerging area of investigation.

In addition to immunotherapy, significant advances in the use of NBs, or other nanoscale US contrast agents such as nanovesicles, will be seen in theranostic applications. Recent reports of gene therapy with nanovesicles show much promise in this area. Using NBs as highly focused delivery vehicles or sensitizers for radiation therapy, chemotherapy, and combination therapies promises to impact cancer treatment significantly [120]. Finally, neuromodulation [119] and gene and drug delivery with NBs for the treatment of neurodegenerative diseases will also emerge as significant areas of potential innovation and impact.

Finally, a note about clinical translation of NB-base technologies. Although there has now been a fair amount of preclinical evidence for the potential of CEUS imaging with NBs, the work has been almost exclusively carried out in small rodents and often without appropriate controls. Thus, a significant amount of work remains before NBs can make their way to routine human use. Achieving investigative new drug status with the Food and Drug Administration will require a considerable and concerted effort. Some crucial steps include (a) scaling up preclinical work to larger animals, such as rabbits, dogs, and nonhuman primates to demonstrate NB safety and feasibility of NBs to achieve sufficient contrast at a clinically appropriate penetration depth for extended time periods compared with mice, (b) toxicology studies in standard models carried out under Good Laboratory Practice protocols, and (c) working out challenges with formulation manufacturing, such as efficient scale-up and extended storage. In addition, the introduction of specific ligands for molecular imaging or image-guided

therapy with contrast will both require additional, highly specialized sonication parameters to work in concert with the NBs and image processing algorithms, including augmentation of quantitative parametric analysis of the kinetic data and creation and validation of unique imaging biomarkers enabled by extravascular targeting. Introducing new software and new hardware capable of taking the best advantage of NB technology in a specific application will require multicenter clinical trials and support from the appropriate professional societies to enable broad adoption.

Potential pitfalls of the NB agents in human use must also be considered. Namely, one apparent challenge could be the sensitivity of NB detection at low frequencies which are suitable for deeper targets. However, modern transducer technology has already sought to address challenges of penetration depth, and thus, it is foreseen that the challenge could be relatively straightforward to address. The alternative could be to focus on applications specific to high-frequency US, which is currently already clinically used for 'small parts' imaging applications such as prostate, thyroid, breast, and lymph as well as pediatric applications.

Conclusions

Existing at the interface of imaging physics, colloid science, and nanomedicine, NBs are a deceptively simple system with broad potential to contribute to the field of biomedical US. Recent interest and research in the field have propelled NBs to a stage where they are now poised to be a significant driver of innovation in the US contrast agent field. The successful clinical translation of NB-based contrast agents depends on a better understanding of NB-US interactions and the establishment of a track record showing exceptional safety, low toxicity, and clear imaging benefits in large animal models compared with existing MB agents. This will, in turn, drive exciting emerging applications in cancer diagnosis, molecular imaging, and gene and drug delivery that can help enhance or change the management of many diseases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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